

DEV

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SUMMARY

I am Dev, a B.Tech Computer Science undergraduate with specialization in AI/ML and a software developer focused on Web Development, blockchain infrastructure, Linux desktop internals, and open source software.

Selected as a **Google Summer of Code 2026 Contributor** at **Sugar Labs** for the **GTK3 to GTK4 Linux Desktop Migration Project**. **Accepted Letter**

Also selected as an **LFX Linux Foundation Mentee** at **LFD T Web3j** for Ethereum protocol development and blockchain infrastructure contributions.

OPEN SOURCE EXPERIENCE

Sugar Labs - Core Desktop Environment

GSoC 2026 Contributor & Open Source Contributor

- Worked on GTK4 migration, Wayland compatibility, D-Bus lifecycle recovery, runtime GDK safety guards, clipboard handling, and Linux desktop stability improvements
- Fixed multiple runtime crashes and compatibility issues affecting Sugar desktop components across GTK3/GTK4 and Wayland environments
- **Merged PRs:** #1030, #1059, #1060, #1061, #1063

LFD T Web3j - Ethereum Protocol

LFX Linux Foundation Mentee & Open Source Contributor

- Worked on Ethereum ABI encoding, transaction signing, EIP-7594 blob transaction wrapper (Fusaka hardfork), websocket fallback systems, transaction pipeline correctness, JVM resource handling, and filter lifecycle management inside the core web3j codebase
- Fixed issues involving **EIP-7594 cell proof encoding and blob wrapper serialization (RLP wrapper_version + cell_proofs for Osaka/Fusaka hardfork)**, EIP-4844 blob transaction signing, Utf8String ABI encoding, nested dynamic array decoding, invalid address recovery, transaction receipt processor shutdown, and ABI struct code generation
- **Merged PRs:** #2255, #2254
- **Open PRs:** #2263 (EIP-7594 Blob Wrapper), #2289, #2288, #2287, #2278, #2277, #2276, #2257, #2274, #2273, #2272

OpenSSF / Hyperledger Ecosystem

Open Source Contributor

- Contributed to OpenSSF RSTUF security infrastructure, Hyperledger Cacti, and Hyperledger Identus involving container security, deployment hardening, blockchain tooling, SDK reliability, CI workflows, and infrastructure maintenance
- **Merged PRs:** RSTUF #968 Cacti #4183, #4180 Identus #591
- **Open PRs:** Cacti #4179, Cacti #4176

Chicago Pediatric Cancer Data Commons

Open Source Contributor

- Worked on ML pipeline debugging and clinical trial matching dashboard improvements
- **PR:** #4

PROJECTS

BTC HashFrame — Bitcoin Encoding System

Live Demo

Next.js, TypeScript, Cryptography, Bitcoin Infrastructure

- Built a client-side Bitcoin encoding platform for air-gapped workflows supporting BIP39, BIP21, PSBT payloads, descriptors, and raw transaction QR generation
- Designed zero-knowledge browser architecture with deterministic validation and secure export flows ensuring sensitive wallet data never leaves the local device

NWC Tester — Lightning Integration Platform

Live Demo

Next.js, TypeScript, Lightning Network, JSON-RPC

- Developed a real-time Nostr Wallet Connect (NWC) testing platform with Lightning Network integration, wallet command execution, transaction inspection, and monitoring workflows
- Integrated Bitcoin Connect and Alby SDK for secure wallet connectivity, invoice handling, payment execution, balance lookup, and responsive debugging dashboards

SHINRA LABS — AI Data Platform

Live Demo

React, Supabase, PostgreSQL Distributed Workflow Systems

- Built a production-style AI data labeling and dataset marketplace platform with scalable annotation pipelines, QA review systems, workforce routing, and realtime workflow management
- Integrated Supabase backend infrastructure with authentication, storage, realtime synchronization, and structured company/freelancer dashboards inspired by enterprise AI tooling platforms

TECHNICAL SKILLS

Languages: Java, Kotlin (basic), TypeScript, JavaScript, Python, SQL, Bash

Blockchain: Ethereum, Web3j, JSON-RPC, Smart Contract Interaction

Systems: Linux, Wayland, D-Bus, GTK3/GTK4, Runtime Debugging

Backend / APIs: REST APIs, Node.js, Distributed Systems, Gradle

Frontend: React, Next.js, Tailwind CSS

DevOps: Docker, GitHub Actions, CI/CD, Container Security